Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 2

Date: 17-04-2026

Design and develop a React JS based web application for Ticket Booking of an Internal Department Event.

SDG: SDG 9 – Industry, Innovation and Infrastructure

Presented By:

N.KARTHIK – VTU26057 – 23UECS0385.

Courses Handling Faculty:

Dr. Hafizur Rahman

INTRODUCTION

- Event ticket booking is essential for managing registrations efficiently
- Manual booking systems are time-consuming and error-prone
- This project provides an online platform for booking event tickets
- It allows users to view event details and book tickets easily
- The system updates ticket availability in real-time
- It improves accuracy and reduces manual effort
- Useful for colleges and internal department events
- Built using React JS
- Simplifies event management and enhances user experience

PROBLEM STATEMENT

- Many institutions use manual methods for event registration
- Manual systems lead to errors and data inconsistency
- No real-time tracking of available tickets
- Difficult to manage large number of participants
- Time-consuming and inefficient process
- Lack of centralized system for booking management
- Risk of overbooking or duplicate entries
- Need for an automated and reliable booking system

OBJECTIVE(S)

1
2
3
4
5
6
7
8
9

- To design and develop a web-based ticket booking system
- To provide real-time event details and ticket availability
- To allow users to book tickets easily
- To prevent overbooking of tickets
- To implement proper input validation
- To improve efficiency and reduce manual errors
- To enhance user experience using React JS

PROPOSED SYSTEM

- A web-based application for booking event tickets
- Displays complete event information to users
- Allows users to enter details and book tickets online
- Automatically updates available ticket count
- Prevents booking more tickets than available
- Provides instant booking confirmation
- Implements validation for all input fields
- Improves efficiency, accuracy, and user experience

PROPOSED SYSTEM ARCHITECTURE

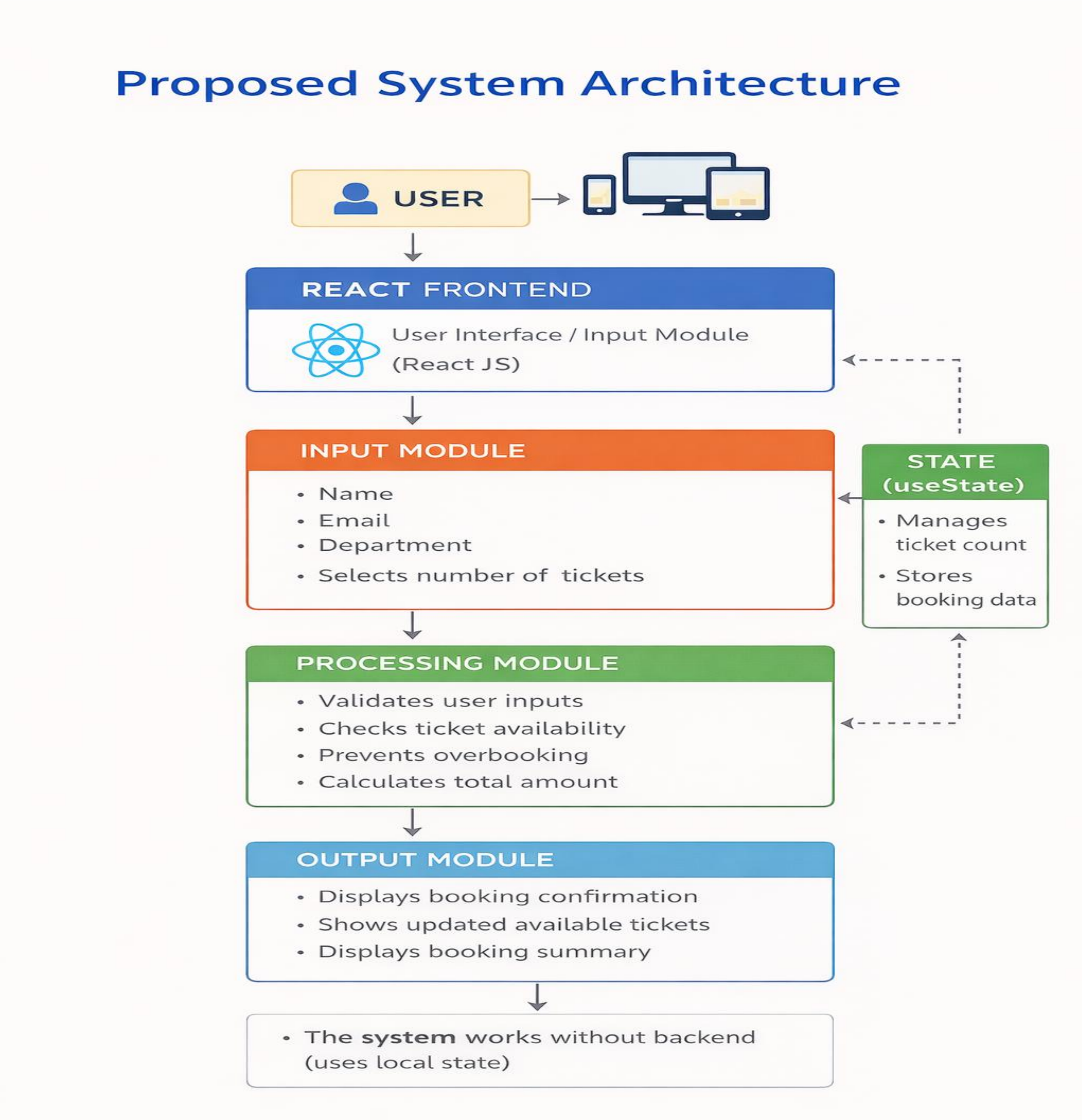
1
2
3
4
5
6
7
8
9

- The system is a web-based React application for ticket booking

- User interacts through the frontend interface (React JS)

- Input Module:
  - User enters name, email, department
  - Selects number of tickets

- Processing Module:
  - Validates user inputs
  - Checks ticket availability
  - Prevents overbooking
  - Calculates total amount



PROPOSED SYSTEM ARCHITECTURE

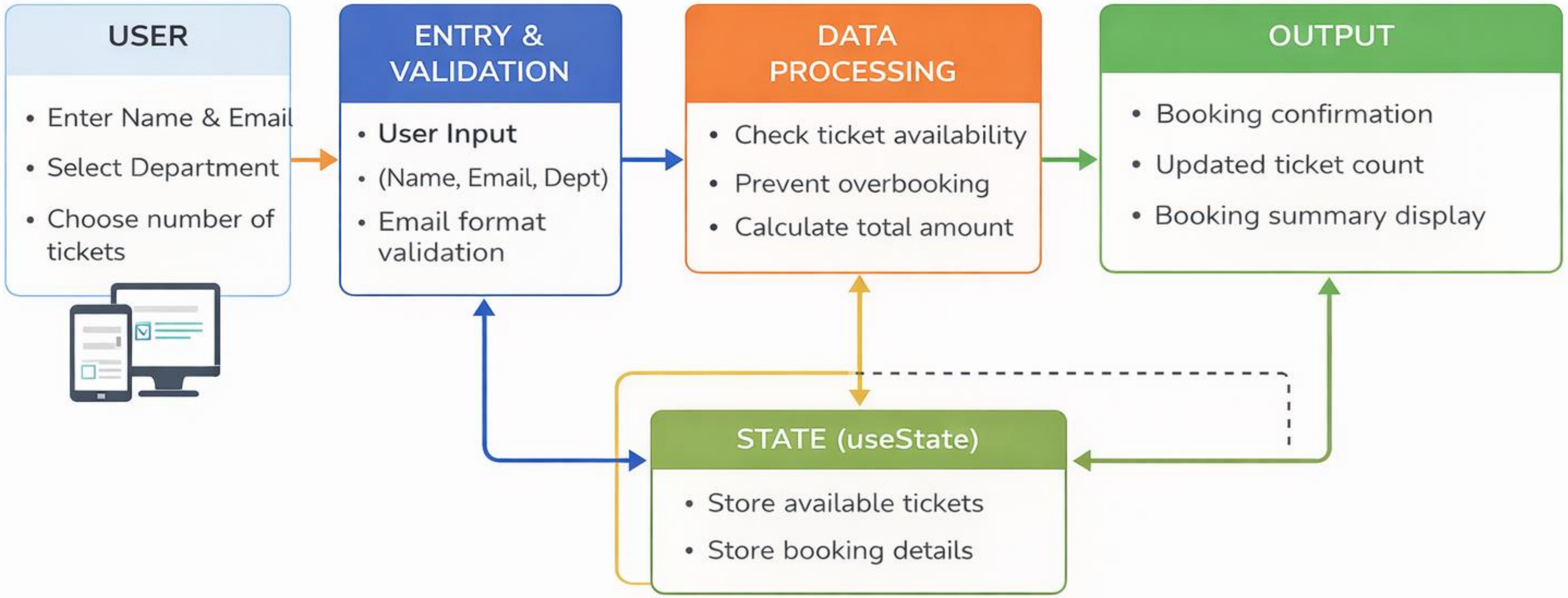
- State Management:
  - useState is used to manage ticket count and booking data

- Output Module:
  - Displays booking confirmation
  - Shows updated available tickets
  - Displays booking summary

- The system works without backend (uses local state)

- Ensures fast performance and user-friendly experience

WORKFLOW


USER
• Enter Name & Email
• Select Department
• Choose number of tickets
ENTRY & VALIDATION
• User Input
• (Name, Email, Dept)
• Email format validation
DATA PROCESSING
• Check ticket availability
• Prevent overbooking
• Calculate total amount
OUTPUT
• Booking confirmation
• Updated ticket count
• Booking summary display
STATE (useState)
• Store available tickets
• Store booking details

TECHNOLOGIES USED

TECHNOLOGIES USED

- Frontend: React JS
- Languages: JavaScript, HTML, CSS
- State Management: useState (React Hook)
- Build Tool: Vite
- Tools: VS Code, Web Browser (Chrome)
- Environment: Node.js & npm
- Operating System: Windows

MODULE DESCRIPTION

Module 1: Event Details Module
- Displays event name, department, date, time, venue
- Shows ticket price and available tickets

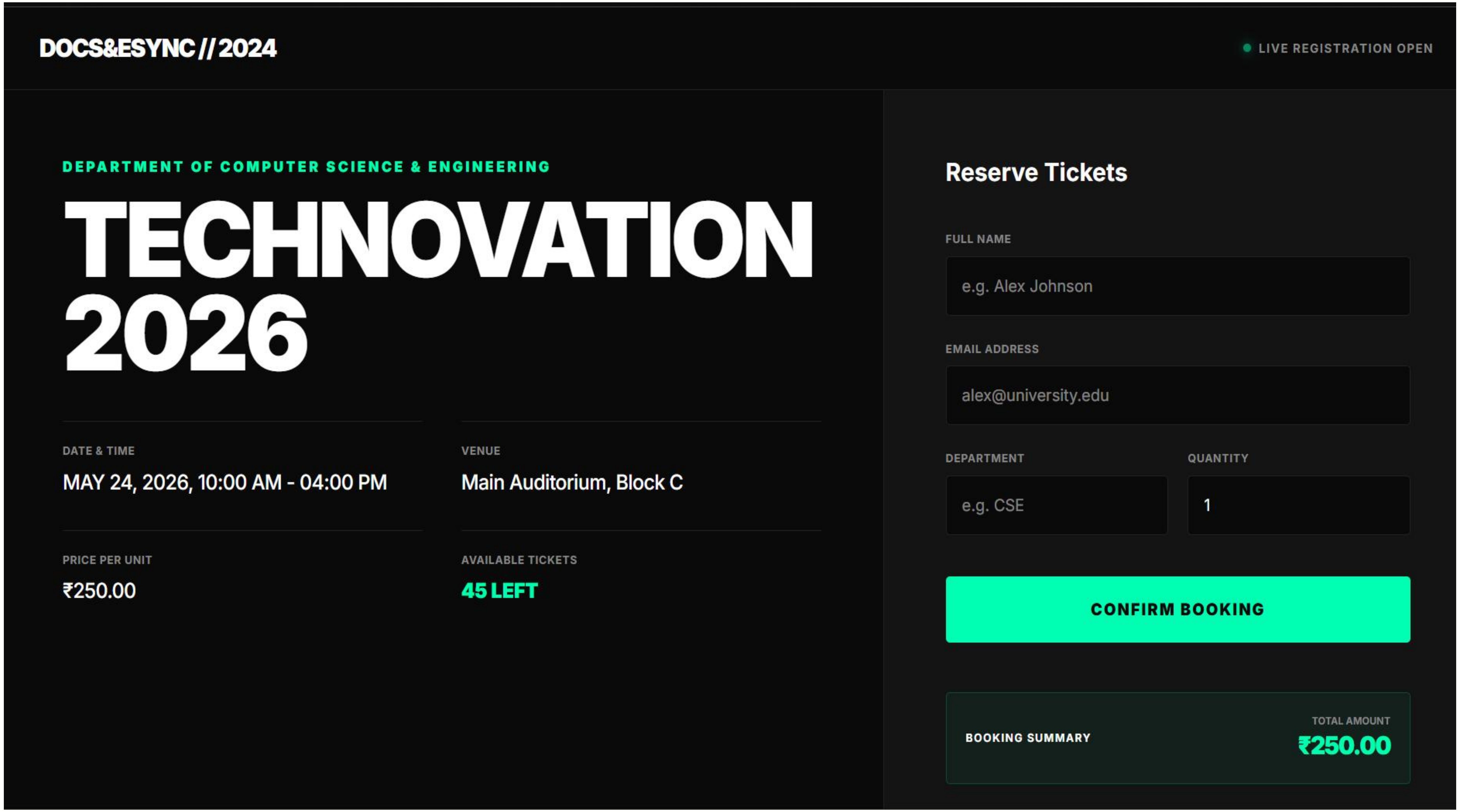
Module 2: Ticket Booking Module
- Allows user to enter name, email, department
- User selects number of tickets

Module 3: Validation Module
- Validates all input fields
- Checks email format
- Ensures ticket count is valid

Module 4: Booking Processing Module
- Checks ticket availability
- Prevents overbooking
- Calculates total amount

Module 5: Output Module
- Displays booking confirmation
- Shows updated ticket count
- Displays booking summary

IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(A)
9

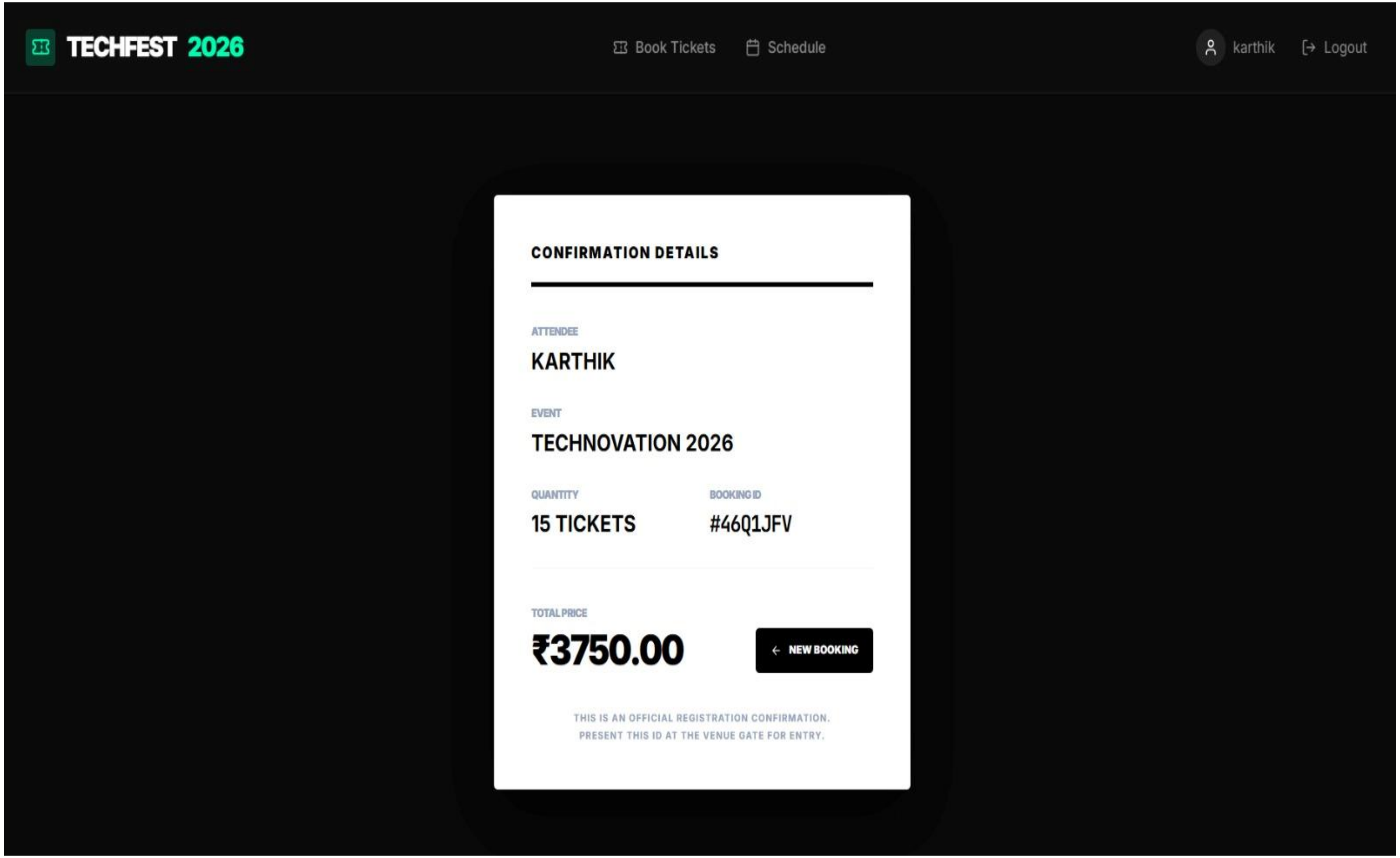
Event Booking Page:

- Displays event details such as event name, department, date, time, and venue
- Shows available number of tickets dynamically
- Displays ticket price information
- Provides a user-friendly booking form

- User can enter:
  - Full Name
  - Email ID
  - Department
  - Number of tickets

- Includes input validation for all fields
- Prevents invalid or empty submissions
- Allows users to confirm registration easily

IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(C)
9

Booking Confirmation Page:

- Displays confirmation message after successful booking
- Shows registration reference ID
- Displays user details:
  - Name
  - Department

- Shows booking details:
  - Number of tickets booked
  - Total amount paid

- Updates available ticket count in real-time
- Provides option to start a new booking session

ADVANTAGES & LIMITATIONS

ADVANTAGES

- Easy and user-friendly interface
- Real-time ticket availability
- Prevents overbooking
- Reduces manual errors
- Fast and efficient booking system

LIMITATIONS

- No backend/database (data not stored permanently)
- No user authentication system
- Limited to small-scale usage
- Requires internet/browser

Thank you